

Patrick Ming

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Education

Washington University in Saint. Louis

Computer Science and Mathematics

Aug 2025 - Exp. Dec 2028

St. Louis, Missouri

- **Relevant Coursework:** MATH 4502: Topics in Applied Math, Machine Learning Methods in Scientific Computing; MATH 4301: Linear Algebra; MATH 4101: Introduction to Analysis; MATH 3300: Matrix Algebra; MATH 3010: Foundations for Higher Mathematics; MATH 2130: Calculus III; CSE2407: Data Structures and Algorithms; CSE 2400: Logic and Discrete Mathematics; SDS 3020: Elementary to Intermediate Statistics and Data Analysis (GPA 4.0/4.0 on Mentioned Coursework)

Experience

Qognitive Inc.

Deep Reinforcement Learning Researcher

May 2026 - July 2026

New York City, New York

- Independently designed a hybrid algorithm for Reinforcement Learning (RL) problems that outperforms SAC and PPO on 9/11 Gymnasium environments tested.
- Built an RL-guided Agentic AI Quantum Algorithmic Cooling controller under a U.S. Department of Energy-funded project with novel hybrid.
- Experimented with Energy-Based Models in RL and developed novel Modular Energy Critic architecture in an attempt to combat Primacy Bias

WashU in St. Louis, Math Department

Teaching Assistant/Grader

Jan 2026 - May 2026

St Louis, MO

- Grader for MATH 3300: Matrix Algebra, SP26.
- Created weekly problem solution manuals, answered questions/requests from email.

Projects

RL Environment for Mathematical Construction/Counterexample Tasks

Post-Training, Reinforcement Learning, LLM Evaluation

Jun 2026 - Now

On Github

- Built an 86-task RL environment and benchmark for math construction/counterexample problems (LinAlg, Algebra, Analysis). Scored with 28 hand-designed verifiers (no LLM judge)
- Red-teamed the verifier for reward-hacker with 735 wrong-object acceptances. Frontier Model (MiniMax-M2) scored 79% correct on the hardest tier.

WUSTL Research Matchmaker

AI Research Assistant + Faculty Research Matchmaker

Jan 2026 - April 2026

On Github

- 3 hr process turned into 5 minutes. Developed a multi-vector RAG model for the purposes of streamlining the research discovery process. Returns a list of compatibility scores with WUSTL faculty based on user prompt.
- Tabulated 1,584 STEM faculty and their research profile using a custom WUSTL webscraper. Website to be made accessible to all first-year STEM courses.

Honors

2024 United States National Chemistry Olympiad (USNCO)

Honors List Award (t150 scorer in the nation)

May 2024

National Level

2024 Mathworks Math Modeling (M3) Challenge

Semi-finalists (top 1-2% of 600 teams)

Mar 2024

National/International Level

Skills and Interests

Languages: Python (NumPy, PyTorch, Pandas, Matplotlib, Flet, CustomTkinter), R, Java, LaTeX

Technologies: Excel, Overleaf, VS Code, Pycharm

Research Interests: Post-Training, Deep Reinforcement Learning, Deep Learning, Machine Learning in Applied Fields, Retrieval Augmented Generation (RAG), Computer Vision, Applied Computational Techniques in Finance.